Below is a question from an user and a candidate response. Please grade the response on a 5-point scale using the following criteria:

1: It means the answer is incomplete, vague, off-topic, controversial, or not exactly what the user asked for. For example, some content seems missing, numbered list does not start from the beginning, the opening sentence repeats user's question. Or the response is from another person's perspective with their personal experience (e.g. taken from blog posts), or looks like an answer from a forum. Or it contains promotional text, navigation text, or other irrelevant information.
2: It means the answer addresses most of the asks from the user. It does not directly address the user's question. For example, it only provides a high-level methodology instead of the exact solution to user's question.
3: It means the answer is helpful but not written by an AI Assistant. It addresses all the basic asks from the user. It is complete and self contained with the drawback that the response is not written from an AI assistant's perspective, but from other people's perspective. The content looks like an excerpt from a blog post, web page, or web search results. For example, it contains personal experience or opinion, mentions comments section, or share on social media, etc.
4: It means the answer is written from an AI assistant's perspective with a clear focus of addressing the instruction. It provide a complete, clear, and comprehensive response to user's question or instruction without missing or irrelevant information. It is well organized, self-contained, and written in a helpful tone. It has minor room for improvement, e.g. more concise and focused.
5: It means it is a perfect answer from an AI Assistant. It has a clear focus on being a helpful AI Assistant, where the response looks like intentionally written to address the user's question or instruction without any irrelevant sentences. The answer provides high quality content, demonstrating expert knowledge in the area, is very well written, logical, easy-to-follow, engaging and insightful.

User: <INSTRUCTION_HERE>

<response><RESPONSE_HERE></response>

Please first briefly describe your reasoning (in less than 100 words), and then write "Score: <rating>" in the last line. Answer in the style of an AI Assistant, with knowledge from web search if needed. To derive the final score based on the criteria, let's think step-by-step.

Figure 7: LLM-as-a-Judge prompt taken from Li et al. [2024].

M_1' : Initialized with M_0, then fine-tuned on the IFT seed data only using SFT.

M_2' : Initialized with M_1', then trained with AIFT(M_1') data using DPO.

M_3' : Initialized with M_2', then trained with AIFT(M_2') data using DPO.

Since we did not use EFT data to train the series of models, they were not always able to score the responses according to the format and even when they did, the scores given typically converged to 4. Therefore, even when starting from the same number of generated new prompts, we could only collect a very small number of valid training samples for DPO. In total, we collected 541 pairs to form the AIFT(M_1') dataset used to train M_2' via DPO, and 429 pairs to form AIFT(M_2') used to train M_3'. The win rates are shown in Figure 8. From the figure we can conclude that EFT data helps to get better performance in the same number of iterations and the gap in performance between the model trained with EFT data and the model trained without EFT data widens in the later iterations.

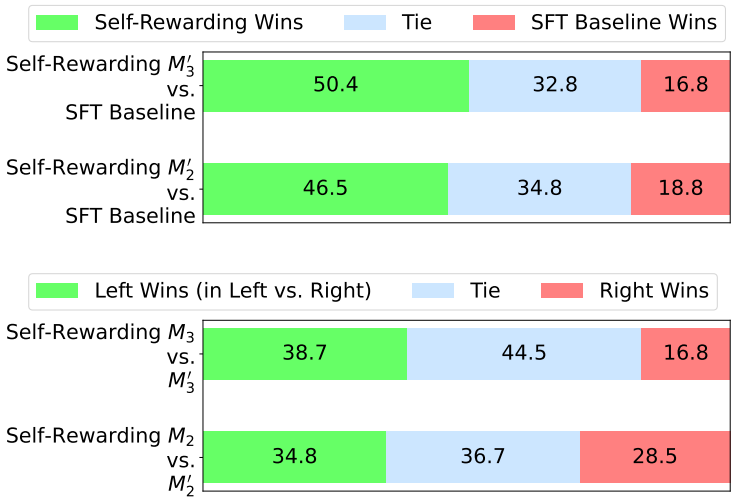

Figure 8: **EFT data helps the self-rewarding loop:** We evaluated the series of models trained using self-reward loops starting from the model trained using only IFT data. We performed head-to-head win rates comparisons on the IFT test set. While M_2' can improve over the SFT baseline and M_3' can improve even more over the SFT baseline, they lag far behind the corresponding models (M_2, M_3) that started from a base model trained using both IFT and EFT data, see Figure 3.

<LIST ALL ALPACAEVAL INSTRUCTIONS>
Given the above list of possible instructions, define a maximum of 20 categories that would cover the types of intructions, for example recipes, reasoning tasks, general knowledge etc. Try to cover as many of the instructions as possible with the maximum 20 categories, while keeping the categories high-level, simple and easy to understand.

Figure 9: Prompt used to obtain instruction categories on the AlpacaEval test set.

Instruction: <INSTRUCTION>
Given the above, categorize it into one of the following 20 categories:
<LIST ALL CATEGORIES>

Secondly, score the instruction in terms of complexity: how complex you think it is to answer from 1-10 (where 10 is a complex question whereby first reasoning or breaking down the question into multiple subquestions for example might help improve the answer).

Thirdly, indicate how long you think the response to the instruction should be, either (a) 1 sentence, (b) 1-3 sentences, (c) 1 paragraph, (d) 2 paragraphs, or (e) 3 or more paragraphs.

Provide your final response in the following format:
Category: <one of the 20 categories>
Complexity: <score out of 10>
Length: <length category>. Do not provide the actual response.

Figure 10: Prompt for categorizing instructions based on their topics, complexities and expected response lengths.

Table 6: Breakdown of AlpacaEval test set instructions by instruction category.

Category	**Number**	**Percentage**
Science / Technology / Engineering	134	16.65%
Professional / Business / Marketing	77	9.57%
Social Interaction / Relationships / Human Behavior	68	8.45%
Miscellaneous / Other	61	7.58%
Mathematics / Logical Reasoning	52	6.46%
Cooking / Recipes	48	5.96%
Software Development / Coding / Algorithms	44	5.47%
Travel / Geography / Exploration	41	5.09%
Literature / Writing / Communication	39	4.84%
History / Social Studies	38	4.72%
Entertainment / Media Analysis	34	4.22%
Language Learning / Linguistics	32	3.98%
Music / Audio / Arts	30	3.73%
DIY Projects / Hobbies	24	2.98%
Technology / Gadgets / Consumer Products	20	2.48%
Gaming / Game Development	18	2.24%
Exercise / Health / Wellness	16	1.99%
Philosophy / Ethics / Ideology	15	1.86%
Sports / Athletics / Physical Activity	12	1.49%
Strategy / Problem-Solving / Critical Thinking	2	0.24%

Table 7: Breakdown of AlpacaEval test set instructions by instruction complexity. The instructions increase in complexity from 1 to 9, where 10 is a complex question that requires first reasoning or breaking the problem into sub-problems before it can be solved.

Complexity	**Number**	**Percentage**
3	238	29.57%
2	206	25.59%
4	122	15.16%
6	79	9.81%
5	68	8.45%
7	41	5.09%
1	34	4.22%
8	14	1.74%
9	3	0.37%

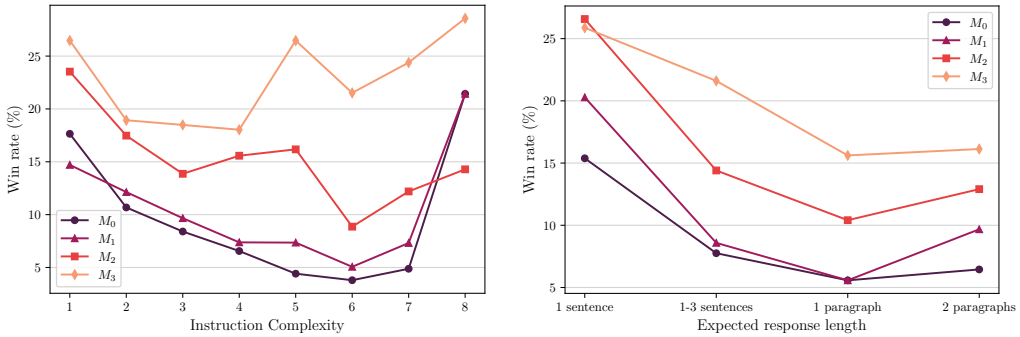

Figure 11: AlpacaEval win rate breakdown for instruction complexities (left) and expected response lengths (right). Self-Rewarding models give gains across most complexities and all response length ranges.